

MODULE 11 · DOMAIN 3 - CLOUD TECHNOLOGY AND SERVICES

# Deploying and Operating on AWS

Four ways to talk to AWS, and two ways to describe what you want built.

DOMAIN 3 of 4   EXAM WEIGHT 34%   MODULE 11 of 26

There is only **one** AWS API. Everything else is a different way of calling it.

# Four ways to interact

| INTERFACE                  | SUITED TO                                                       |
|----------------------------|-----------------------------------------------------------------|
| AWS Management Console     | Exploring, learning, one-off tasks, visual review               |
| AWS Command Line Interface | Scripting, automation, repeatable operations from a terminal    |
| AWS SDKs                   | Calling AWS from inside your own application code               |
| Infrastructure as code     | Describing a whole environment so it can be rebuilt identically |

## ON THE EXAM

Every one of these produces the same API calls, and CloudTrail records all of them the same way.

# Console or command line?

## MANAGEMENT CONSOLE

### A person, doing something once

- Point and click, no syntax to learn
- Good for investigating and for learning
- Hard to repeat identically
- No record of the steps taken

## AWS CLI

### A script, doing something repeatedly

- Commands can be saved and re-run
- Works in automation and pipelines
- Identical every time it runs
- Reviewable, because it is text

## THE TIEBREAKER

One-off or exploratory means the Console. Repeatable, scriptable or automated means the CLI.

# SDKs and the API

- ◆ **An SDK is a library for a programming language** that wraps the AWS API.
- ◆ Available for Python, Java, JavaScript, .NET, Go, and others.
- ◆ Use one when your application itself must call AWS, rather than a person or a script.
- ◆ **Credentials still apply.** Code running on AWS should use an IAM role, never embedded keys.
- ◆ You are not expected to write code on this exam. Recognising what an SDK is for is enough.

## ON THE EXAM

A person means the Console. A script means the CLI. An application means an SDK.

Infrastructure as code turns an environment into a **document you can review, version and rebuild.**

# AWS CloudFormation

- ◆ **You write a template** in JSON or YAML describing the resources you want.
- ◆ **CloudFormation creates them as a stack**, in the right order, working out dependencies for you.
- ◆ Deleting the stack removes everything it created, which makes test environments cheap and safe.
- ◆ Drift detection reports resources that have been changed outside the template.
- ◆ The service is free. You pay only for the resources it creates.

## ON THE EXAM

Provision the same environment repeatedly and identically is the CloudFormation signature phrase.

# Why infrastructure as code wins marks

- ◆ **Repeatability.** The same template produces the same environment every time.
- ◆ **Version control.** Infrastructure changes get reviewed like application changes.
- ◆ Consistency across environments, so development matches production.
- ◆ Faster recovery, because rebuilding is running a template rather than remembering what you did.
- ◆ It removes human error from provisioning. That is usually the phrase in the correct answer.

## ON THE EXAM

Any scenario about inconsistent environments, manual mistakes, or rebuilding quickly points at CloudFormation.

# CloudFormation or the CDK?

## AWS CLOUDFORMATION

### Declarative template

- You write JSON or YAML
- Describes the end state directly
- No programming knowledge needed
- The engine that actually builds things

## AWS CDK

### Familiar programming language

- You write TypeScript, Python, Java and others
- Loops and conditions instead of repetition
- Synthesises into a CloudFormation template
- CloudFormation still does the work

## THE TIEBREAKER

The CDK does not replace CloudFormation. It generates CloudFormation, which then provisions the resources.

# AWS Elastic Beanstalk

- ◆ **You provide application code. AWS provisions and runs the environment.**
- ◆ It creates EC2 instances, a load balancer, Auto Scaling and health monitoring for you.
- ◆ Supports Java, .NET, PHP, Node.js, Python, Ruby, Go and Docker.
- ◆ **You keep full access to the underlying resources** and can adjust them if you need to.
- ◆ The service is free. You pay for the EC2 instances and other resources it creates.

## ON THE EXAM

Deploy an application without managing the infrastructure is the Beanstalk phrase.  
Note it still uses EC2 underneath.

# CloudFormation or Elastic Beanstalk?

## AWS CLOUDFORMATION

### You describe the infrastructure

- Any AWS resource, in any arrangement
- You decide the architecture
- Input is a template
- Complete control, more work

## AWS ELASTIC BEANSTALK

### You provide the application

- A standard web application environment
- AWS decides the architecture
- Input is your code
- Fast, opinionated, less control

## THE TIEBREAKER

Define any infrastructure as code means CloudFormation. Upload an app and let AWS run it means Beanstalk.

# The three deployment models

| MODEL       | WHAT IT MEANS                                                        |
|-------------|----------------------------------------------------------------------|
| Cloud       | Everything runs in the cloud. Built for the cloud or fully migrated. |
| Hybrid      | Some workloads in the cloud, some on premises, connected together.   |
| On-premises | Everything in your own data centre. Sometimes called private cloud.  |

## ON THE EXAM

Hybrid is the one that appears most, because regulation and legacy systems keep some workloads at home.

# Connecting a hybrid environment

- ◆ **AWS Site-to-Site VPN** connects your network to a VPC over the internet, encrypted.
- ◆ **AWS Direct Connect** provides a dedicated private link with consistent bandwidth.
- ◆ **AWS Outposts** puts AWS hardware in your own data centre.
- ◆ AWS Storage Gateway gives on-premises systems access to AWS storage.
- ◆ Module 15 covers the networking of these properly. This is the map.

## ON THE EXAM

Quick and cheap over the internet means VPN. Consistent and private means Direct Connect.

A company creates a test environment by hand in the Console before each release. Engineers report that the test environment frequently differs from production in small ways, and that a recent outage was traced to a security group rule present in one but not the other.

The company wants test environments that match production exactly and can be destroyed afterwards.

**Which service should the company use?**

- |                                                                            |                                                  |
|----------------------------------------------------------------------------|--------------------------------------------------|
| <input type="checkbox"/> A AWS CloudFormation                              | <input type="checkbox"/> B AWS Elastic Beanstalk |
| <input type="checkbox"/> C The AWS CLI, with a documented list of commands | <input type="checkbox"/> D AWS Config            |

◆ PAUSE AND DECIDE

## Which service should the company use?

- A

**AWS CloudFormation**  
One template builds both environments identically, and deleting the stack removes everything cleanly.
- B

**AWS Elastic Beanstalk**  
*Runs an application environment, but the problem is the whole infrastructure matching, not deploying an app.*
- C

**The AWS CLI, with a documented list of commands**  
*Better than clicking, but a human still runs the list and can still skip a step. Documentation drifts.*
- D

**AWS Config**  
*Would detect that the environments differ. Detecting the drift is not the same as preventing it.*

### THE SIGNAL

Match exactly and be destroyed afterwards are both CloudFormation stack properties.

A small team has written a Python web application. They want it running on AWS with a load balancer and automatic scaling, but nobody on the team has infrastructure experience and they do not want to design an architecture.

They do want the ability to adjust the instance type later if the application needs it.

### Which service should the team use?

- |                                                                                |                                                  |
|--------------------------------------------------------------------------------|--------------------------------------------------|
| <input type="checkbox"/> A AWS CloudFormation                                  | <input type="checkbox"/> B AWS Elastic Beanstalk |
| <input type="checkbox"/> C Amazon EC2 with a manually configured load balancer | <input type="checkbox"/> D AWS Lambda            |

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#### ◆ PAUSE AND DECIDE

## Which service should the team use?

- A

**AWS CloudFormation**  
*Requires them to describe the architecture, which is exactly what they said they cannot do.*
- B

**AWS Elastic Beanstalk**  
*Upload the code; AWS builds the environment. And the underlying resources stay adjustable.*
- C

**Amazon EC2 with a manually configured load balancer**  
*The most infrastructure work of all four, for a team that has none of that experience.*
- D

**AWS Lambda**  
*Serverless would remove infrastructure entirely, but it also removes the instance type they want to control.*

### THE SIGNAL

Do not want to design an architecture rules out CloudFormation. Adjust the instance type later rules out Lambda.

An operations engineer needs to stop three hundred EC2 instances across several accounts at the end of each working day, and start them again each morning. The instance identifiers are known and stable.

The task must run without anyone being present.

### Which approach is most appropriate?

- ☐ A Use the AWS Management Console each morning and evening
- ☐ B Use the AWS CLI in a scheduled script
- ☐ C Write a CloudFormation template
- ☐ D Deploy the instances with Elastic Beanstalk

### ◆ PAUSE AND DECIDE

## Which approach is most appropriate?

- A

 Use the AWS Management Console each morning and evening  
*Requires a person twice a day, which the last sentence explicitly forbids.*
- B

 Use the AWS CLI in a scheduled script  
*Scriptable, repeatable and runs unattended. Exactly what the CLI exists for.*
- C

 Write a CloudFormation template  
*Describes what infrastructure should exist. It is not a tool for starting and stopping existing instances on a schedule.*
- D

 Deploy the instances with Elastic Beanstalk  
*Re-deploys the workload to solve a scheduling problem. Nothing about the application needs changing.*

### THE SIGNAL

Without anyone being present is the automation requirement. Repeated operations on existing resources means the CLI.

A manufacturer is moving most workloads to AWS, but a regulator requires its quality-records system to remain in its own data centre for at least three more years. That system must exchange data continuously with applications now running in AWS.

The company wants consistent, predictable network performance between the two.

**Which deployment model and connection does this describe?**

- |                                                                     |                                                                      |
|---------------------------------------------------------------------|----------------------------------------------------------------------|
| <input type="checkbox"/> A Cloud deployment with Amazon CloudFront  | <input type="checkbox"/> B Hybrid deployment with AWS Direct Connect |
| <input type="checkbox"/> C On-premises deployment with AWS Outposts | <input type="checkbox"/> D Hybrid deployment with a Site-to-Site VPN |

◆ PAUSE AND DECIDE

## Which deployment model and connection does this describe?

- A

 Cloud deployment with Amazon CloudFront  
*Not cloud deployment; one system stays on premises by law. CloudFront caches content and is irrelevant here.*
- B

 Hybrid deployment with AWS Direct Connect  
*Workloads in both places, and Direct Connect gives the consistent, predictable performance asked for.*
- C

 On-premises deployment with AWS Outposts  
*Most workloads are moving to AWS, so this is not on-premises. Outposts would also not satisfy the regulator's system staying put.*
- D

 Hybrid deployment with a Site-to-Site VPN  
*Correct model, wrong connection. A VPN runs over the internet, so performance varies.*

### THE SIGNAL

Two halves to get right. Workloads in both places is hybrid; consistent and predictable is Direct Connect, not VPN.

# What to carry out of this module

- ◆ **One API, four interfaces.** A person means Console, a script means CLI, an application means an SDK.
- ◆ **CloudFormation describes infrastructure. Beanstalk runs your application.**
- ◆ The CDK does not replace CloudFormation. It generates it.
- ◆ Repeatable, identical, no human error and destroy afterwards are all CloudFormation phrases.
- ◆ Describing infrastructure and operating existing infrastructure are different jobs.

## ON THE EXAM

Hybrid plus consistent performance means Direct Connect. Hybrid plus quick and cheap means Site-to-Site VPN.